



CS 540 Module Six Activity Guidelines and Rubric

Overview

Understanding how multi-level data modeling improves visualization effectiveness is a critical skill for working with complex data sets. Communicating modeling decisions clearly ensures that visualizations convey accurate insights and support understanding across technical and nontechnical audiences.

In academic and professional environments, you will need to justify how data is structured and presented. Clear explanations of modeling choices strengthen your ability to contribute to data-driven conversations, create meaningful visual stories, and support decisions with well-structured evidence.

Directions

In this activity, you will design and implement a Python-based dynamic data visualization dashboard. The goal is to apply principles of data visualization design, utilize advanced Python visualization libraries, and implement multi-level exploration techniques with interactive elements. You will also customize the visualizations to enhance usability, accessibility, and aesthetic appeal. You will use your development environment to complete this activity. When you are ready to begin your work, access Codio by clicking the title of the next task.

In the Jupyter Notebook, complete the following steps:

- Write the necessary code in the "To Do;" placeholder code blocks. Use the comments to guide your code.
- Run each code block successfully before doing the next.
- At the end, restart Kernel and run all the code end-to-end to generate all visuals.
- Save the visuals in PNG file formats, ensuring to add calls to display the images as output.

When the activity is complete, download the HTML version of your project by clicking on **File**, then **Download As**, then **HTML (.html)** from the Jupyter menu bar. Save the file to your local machine.

***Note:** if the code block was not run in the Notebook, it may not render in the HTML output of the Notebook. To fix missing blocks, run the code block, then download as HTML.*

Specifically, you must address the following rubric criteria:

Part One: Data Set Selection and Preparation

1. **Preprocess** the data set using Python. Consider the following:

- A. Are there missing values in the data set?
- B. Are there numerical values that need to be normalized or scaled to ensure accurate comparison and visualization?

Part Two: Static Visualization with Design Principles

2. Create multiple **static visualizations** using Matplotlib or Seaborn to highlight key insights from the data set. Consider the following:

- A. Heatmap showing correlations between variables
- B. Hierarchical bar chart comparing categories and subcategories
- C. Time series line chart with annotations

3. Apply **visualization design principles**. Include the following:

- A. Use effective color palettes
- B. Label axes, titles, and legends clearly
- C. Adjust scale and proportions for clarity

Part Three: Interactive Visualizations

4. Create multiple **interactive visualizations** using Plotly. Include the following:

- A. A visualization that uses a drill-down or zoom feature
- B. Another visualization using Overview+Detail

5. Add **interactive elements** to your visualizations. Consider the following:

- A. Tooltips that reveal additional information on hover
- B. Dropdown menus or buttons to filter or control the data display

Part Four: Customization and Animation

6. **Customize** your visualizations. Include the following:

- A. A consistent visual theme (fonts, colors, gridlines)
- B. An optimized layout for usability and clarity

7. Add **animation and transition effects** to your interactive visualizations.

Part Five: Reflection

8. Explain how specific **design choices** contributed to the clarity and impact of the visualizations.

9. Describe any **challenges** you faced when implementing interactivity and animation. Include the following:

- A. How you addressed each challenge

10. Evaluate how **accessibility and cognitive load** considerations were incorporated into the design of the visualizations.

What to Submit

To complete this activity, you must submit the following:

Part One through Part Four

Download your Jupyter Notebook as an HTML file. Review the file to ensure the code ran successfully and all visualizations are displayed. Submit the HTML file.

Part Five

Submit your reflection as a 2-to-3-page Microsoft Word document with double-spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module Six Activity Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Preprocess	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Preprocesses the data set using Python	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Static Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Creates multiple static visualizations using Matplotlib or Seaborn to highlight key insights from the data set	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Visualization Design Principles	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Applies visualization design principles, including effective color palettes, axes, titles, and legends labeled clearly, scale and proportions adjusted for clarity	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Interactive Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Creates multiple interactive visualizations using Plotly, including a visualization that uses a drill-down or zoom feature, and another visualization using Overview+Detail	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Interactive Elements	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Adds interactive elements to the visualizations	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Customizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Customizes the visualizations, including a consistent visual theme (fonts, colors, gridlines) and an optimized layout for usability and clarity	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Animation and Transition Effects	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Adds animation and transition effects to the interactive visualizations	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Design Choices	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains how specific design choices contributed to the clarity and impact of the visualizations	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Challenges	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes any challenges faced when implementing interactivity and animation, including how each challenge was addressed	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Accessibility and Cognitive Load	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Evaluates how accessibility and cognitive load considerations were incorporated into the design of the visualizations	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	9.5
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
Total:					100%